

VOID-NAV: Finding Your Way When GPS Goes Dark

A low-cost navigation box that works out where it is from the Earth's magnetic field, using quantum algorithms to match what it senses against a map. Built from scratch in 24 hours.

SENSE

9-axis IMU + magnetometer, 50 Hz

MATCH

Grover's search on a magnetic map

CLASSIFY

Quantum-kernel SVM identifies the zone

FUSE

Particle filter gives a live position

Team VOID

Without GPS, every navigation system slowly gets lost, and nobody can tell by how much.

GPS is fragile

Satellite signals are extremely weak by the time they reach the ground. Interference, terrain, tunnels, bunkers, dense forest and underwater operations all cut them off.

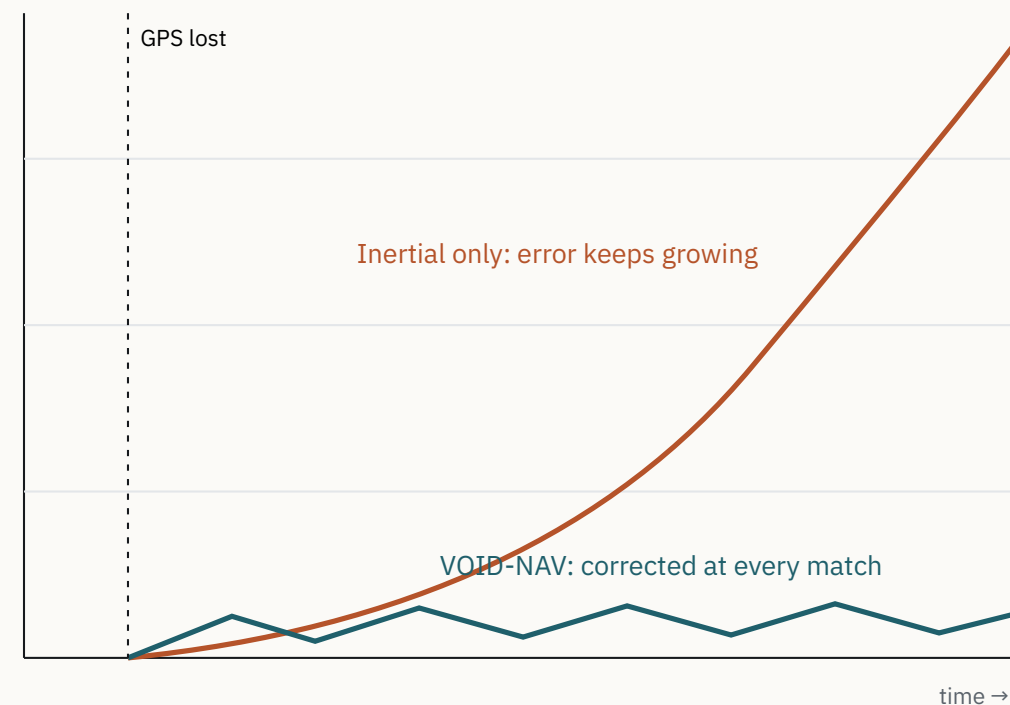
The backup drifts

Inertial sensors (accelerometers and gyros) estimate motion, but small errors add up. A low-cost unit can be off by **hundreds of metres within minutes**.

Good alternatives are expensive

Navigation-grade inertial units and emerging quantum inertial sensors cost lakhs to crores and are export-controlled.

POSITION ERROR AFTER GPS IS LOST (ILLUSTRATIVE)



What we are solving.

“Give soldiers, vehicles and drones a position they can trust when GPS is unavailable, using a sensor that emits nothing, costs under ₹5,000, and is ready to take a quantum magnetometer when one is available.”

Who needs it

Infantry in bunkers and tunnels, armoured vehicles, UAVs, border patrol teams.

What they need

A position within a few metres, a confidence level, and no radio emissions.

Why today fails

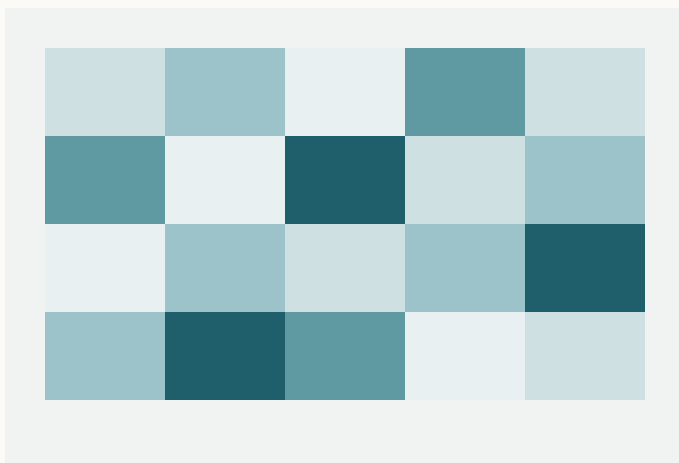
Low-cost inertial units drift; good ones are too costly to issue widely.

Use case fit

UC-085: quantum sensing for GPS-denied navigation, aligned with the NQM sensing hub.

Every place has its own magnetic fingerprint. We map it once, then navigate by it.

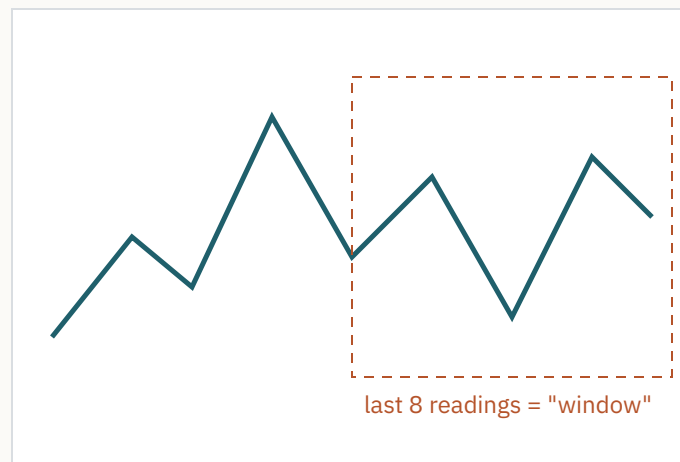
STEP 1 · MAP (ONCE)



Walk the area with GPS on

Log the field strength per 2 m grid cell, building a "magnetic map"

STEP 2 · SENSE (GPS OFF)



Move through the area

IMU estimates motion, magnetometer records a short field profile

STEP 3 · MATCH + CORRECT

QUANTUM

Grover's search finds map cells matching the window

QSVM confirms which zone the device is in

CLASSICAL

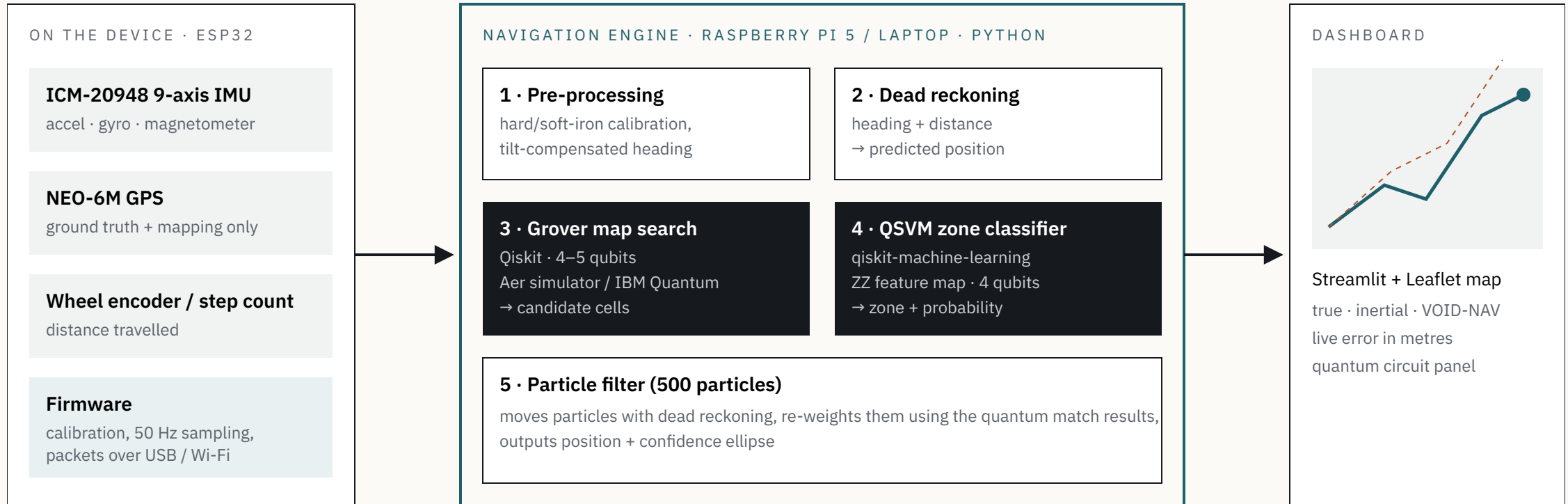
Particle filter snaps the drifting track back

Output position + confidence

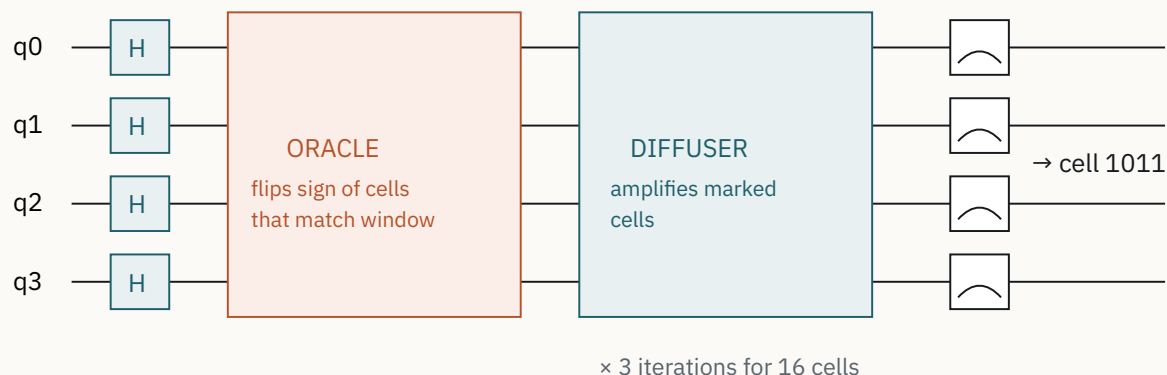
Shown live on a map dashboard; nothing is transmitted

Indoors and near structures, steel and wiring make the field vary strongly over a few metres, which is ideal for a campus demo. Outdoors, geological anomaly maps play the same role.

How the pieces connect.



Grover's algorithm finds the matching spot on the map in about $\sqrt{N}$ steps instead of N .



MEASURED PROBABILITY PER CELL (SIMULATOR, EXAMPLE)



In plain terms

The map is split into 16 cells (4 qubits). The oracle "marks" every cell whose stored field values match what the sensor just measured, within a tolerance. Grover then boosts the marked cells so that measuring the qubits gives the right cell with high probability.

Why it matters

A classical search checks every cell. Grover needs about $\sqrt{N}$ checks: 1 million cells → about 1,000 steps. That is what larger, country-scale maps will need.

Source: adapted from [qiskit-community-tutorials/algorithms \(Grover\)](#), ported to current Qiskit 1.x. At hackathon scale the oracle is built from a classical comparison, so the speed-up is shown in principle, not measured.

A quantum kernel tells apart magnetic signatures that look alike to classical methods.

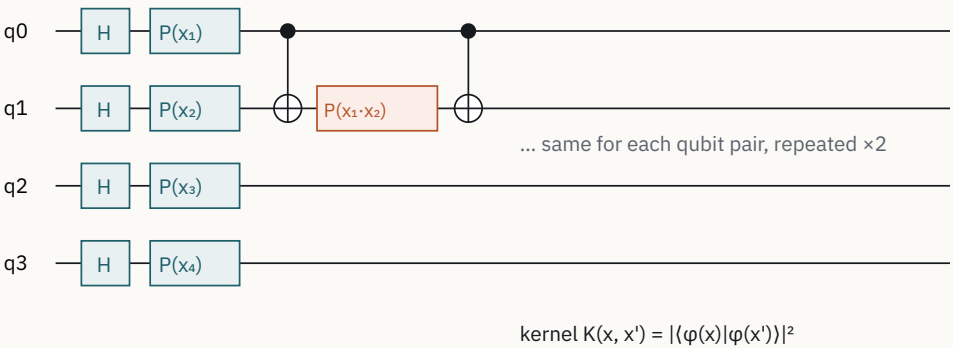
INPUT FEATURES (PER 1-SECOND WINDOW)

x ₁	Total field strength B (μT)
x ₂	Vertical component B _z
x ₃	Change in B over the window
x ₄	Field inclination angle

How it works

Each window of readings is encoded into 4 qubits with a **ZZ feature map**. The overlap between two encoded states is the "similarity" (kernel). A standard SVM uses that kernel to decide which zone of the map, e.g. *corridor A, stairwell, lab*, the device is in. This narrows the search before Grover runs.

ZZ FEATURE MAP · 4 QUBITS · 2 REPS



PLAN	DETAIL
Training data	~300 labelled windows from the mapping walk
Baseline	Classical RBF-kernel SVM on the same data
Report	Accuracy of both, side by side, on held-out data

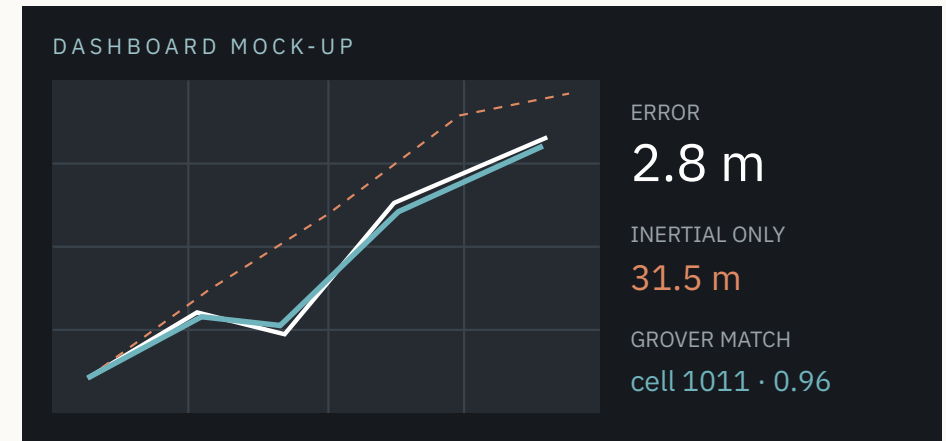
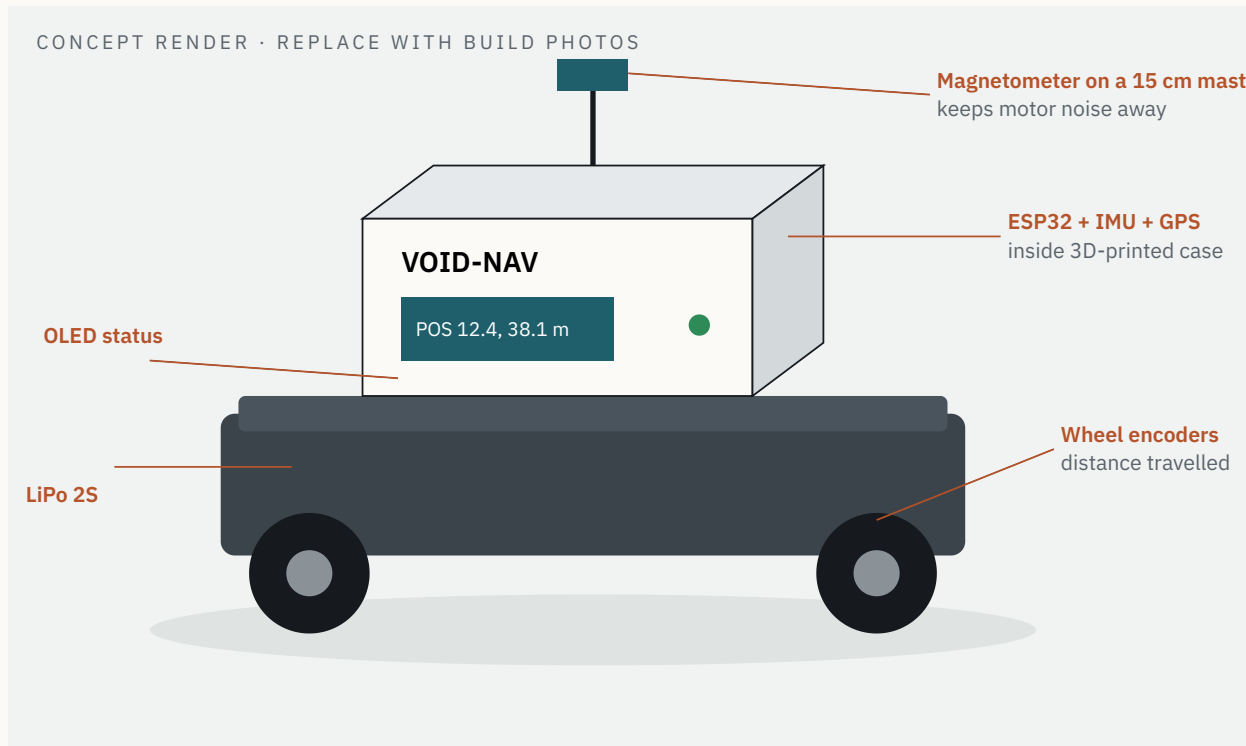
Adapted from the QSVM tutorial in qiskit-community-tutorials; implemented with FidelityQuantumKernel + QSVC.

Every algorithm in the system, and what runs where.

#	COMPONENT	TYPE	WHAT IT DOES	RUNS ON	BUILT IN 24 H?
Q1	Grover's search	Quantum algorithm	Finds map cells matching the live magnetic window	Qiskit Aer; one run on IBM Quantum hardware	Yes
Q2	Quantum-kernel SVM	Quantum machine learning	Classifies which zone of the map the device is in	qiskit-machine-learning (simulator)	Yes
Q3	Quantum magnetometer interface	Quantum sensing (planned)	Drop-in slot for an NV-diamond magnetometer, 1000× more sensitive	Same data format as today's sensor	Interface only
C1	Sensor calibration	Classical	Removes hard/soft-iron distortion, tilt-compensates heading	ESP32 + Python	Yes
C2	Dead reckoning	Classical	Predicts position from heading + distance	Python	Yes
C3	Particle filter	Classical	Fuses prediction with quantum match results into one position	Python / NumPy	Yes

Why a hybrid? Today's quantum computers are small and noisy, so we use them where they fit best: search and pattern comparison. Everything time-critical stays classical, which is how practical quantum systems are being built.

A hand-carried sensor box on a small rover, plus a laptop dashboard.



Numbers in the mock-up are placeholders; real values come from the demo run.

Two ways to carry it

Rover: a 4-wheel RC chassis for a repeatable demo track.

Handheld: the same box in a backpack strap, using step counting instead of wheel encoders, to show the soldier use case.

Everything is available off the shelf in India, for under ₹5,000.

PART	ROLE	APPROX. ₹
ESP32 DevKit V1	Sensor hub, Wi-Fi	450
ICM-20948 9-axis IMU	Accel, gyro, magnetometer	900
NEO-6M GPS module	Mapping + ground truth	450
4WD robot chassis + motors	Rover platform	1,200
Encoders + L298N driver	Distance, motor control	400
0.96" OLED display	On-device status	200
2S LiPo + buck converter	Power	900
Total		≈ 4,500

WIRING (I²C + UART)



A laptop or Raspberry Pi 5 runs the navigation engine and receives data from the ESP32 over USB serial or Wi-Fi (UDP).

Open-source tools a student team already knows.

FIRMWARE

Arduino C++

ESP32 core, SparkFun ICM-20948 library, TinyGPS++, Adafruit SSD1306

QUANTUM

Qiskit 1.x

qiskit-aer for simulation, qiskit-ibm-runtime for real hardware, qiskit-machine-learning for QSVM

NAVIGATION

Python 3.11

NumPy, SciPy, FilterPy (particle filter), pyserial

BASELINES

scikit-learn

Classical RBF-SVM and brute-force search to compare against quantum

DASHBOARD

Streamlit + Folium

Live map with three tracks, error meter, circuit and histogram panels

DATA

CSV + Parquet

Magnetic map grid, labelled training windows, run logs for replay

HARDWARE CAD

Fusion 360 / Tinkercad

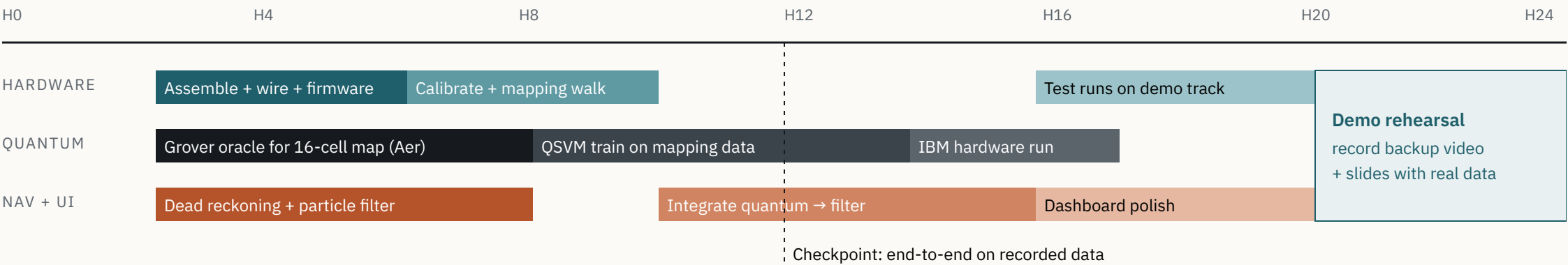
3D-printed sensor case and mast mount

COLLABORATION

GitHub

One repo: /firmware, /quantum, /nav, /dashboard, /data

Our build plan, hour by hour, with a fallback at every step.



RISK	FALLBACK
Magnetometer noise from motors	Mast mount + calibration; handheld mode needs no motors
IBM Quantum queue too long	Run on Aer simulator live; show a pre-run hardware result
Live demo fails on the day	Replay a logged run through the same pipeline + backup video

What the judges will see, and why it is different.

DEMO SCRIPT · 3 MINUTES

1. **Show the map:** the magnetic heat-map of the corridor we recorded.
2. **Switch GPS off** on the dashboard and drive the rover along the track.
3. **Watch the drift:** the inertial-only track wanders off course.
4. **Quantum match:** Grover's histogram spikes on the right cell; QSVM names the zone.
5. **Correction:** the VOID-NAV track snaps back and the error meter drops.
6. **Result:** final error for both methods, shown side by side.

N - 01

Quantum map matching

Grover's search used for navigation map matching, not a textbook example.

N - 02

Two quantum checks, one answer

QSVM narrows the zone, Grover picks the cell, and the filter only trusts them when they agree.

N - 03

Quantum-sensor ready

The sensor interface matches an NV-diamond magnetometer's output, so a quantum sensor can replace today's chip without changing the software.

From a campus corridor to a field-ready defence module.

TODAY · HACKATHON

Working prototype

MEMS magnetometer, 16-cell map, Grover + QSVM on simulator, live dashboard. Target: several times lower error than inertial-only.

NEXT · 6 MONTHS

Larger maps, real terrain

Campus-wide map, outdoor trials using public geomagnetic anomaly data, larger Grover circuits on IBM hardware.

LATER · 18 MONTHS

Quantum sensor upgrade

Swap in an NV-diamond magnetometer with an NQM sensing-hub partner; trials with defence labs on UAVs and vehicles.

<₹5k

prototype cost

0

radio emissions: fully passive

2

quantum algorithms in the loop

24 h

from parts to working demo

SUMMARY

When GPS goes dark, the ground still knows where you are.

VOID-NAV: magnetic-fingerprint navigation with quantum search and quantum machine learning, built by students in 24 hours and ready for a quantum sensor.

Team VOID

THANK YOU · QUESTIONS WELCOME

UC-085 · QUANTUM SENSING · DEFENCE